

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Debugging & Optimization Task

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	Assessment:	Assessment Tool 2 – Debugging & Optimisation Task
Weightage:	35% of CIA	Total Marks:	100
CO2 Statement:	Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.		
Student Name:	P. Harsha vardhan	Reg. No.:	192411179
Section / Batch:	CSE	Date:	22/07/2026

Code of Conduct

I _____ P. Harsha vardhan _____
(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: p h a r s h a

Instructions

- This test contains 20 Debugging & Optimisation Task questions. Total: 100 Marks (20*5).
- When a student answer using Scenario-Based, in evaluation the answer should be with following five requirements: Identify the problem type and constraints, Select appropriate data structure, Design the Solution, Optimization and efficiency, Clarity of representation.
- No negative marking. Unanswered questions carry zero marks.

Section / Topic	Focus	Q Nos	Marks
Search Data Structure, Stack	Binary Search, Stack Operations	1 - 5	25

Linked List Data Structure	List Operations	6 – 10	25
Queue Data Structures	Queue Operations	11 - 15	25
Queue Data Structures, Stack Notations	Queue Operations, Expression Evaluation	16 - 20	25
GRAND TOTAL:		100 Marks	

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Debugging Accuracy	20	All errors corrected; code runs flawlessly	Most errors corrected; minor issues remain	Partial correction; code runs with issues	Code fails or major errors persist
Error Identification	30	Accurately identifies all logical, syntax, and edge-case errors	Identifies most major errors	Identifies some errors but misses key issues	Struggles to identify errors
Code Quality & Readability	20	Clean, modular, well documented, follows best practices	Mostly clean and readable	Some structure but inconsistent	Poorly written, hard to understand
Explanation & Justification	20	Clear, logical, and well justified reasoning with complexity analysis	Good explanation with some justification	Limited explanation; lacks depth	No or incorrect explanation
Testing & Validation	10	Comprehensive test cases including edge cases	Adequate test cases	Minimal testing	No testing provided
Faculty In-charge		Course Coordinator		Program Director	
Signature: _____		Signature: _____		Signature: _____	
Date: _____		Date: _____		Date: _____	

DEBUGGING & OPTIMIZATION TASK (Set 1)



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192411179RB

Q1
Binary Search
5 marks

Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures

What is missing in the below pseudocode?

DEQUEUE(Q):
if front == -1:
 print("Underflow")
else:
 return Q[front++]

✳️ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 DEQUEUE(Q):  
2 if front == -1:  
3   print("Underflow")  
4 else:  
5   return Q[front++]
```

Test Input

stdin input (optional)

Output



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Q2
Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Stack

```
new.next = head  
head = new
```

✳️ **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 new.next = head  
2 head = new
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

Q3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marksQ8
Linked List Data Structures
5 marksQ9
Linked List
5 marks

Q10

current = current.next

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while current != NULL:
2     next_node = current.next
3     current.next = prev
4     prev = current
5     current = next_node
```

Test Input

stdin input (optional)

Output



Run



Save



Submit Fix

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marks

Q8

Q7 Stack

5 marks

Point out the error.

POP(stack):

if top == 0:

print("Underflow")

else:

return stack[top--]

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 def POP(stack, top):
2     if top == -1:
3         print("Underflow")
4         return None
5     else:
6         element = stack[top]
7         top -= 1
8         return element
```

Test Input

stdin input (optional)

Output



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Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack 5 marks

Q6 Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List Stack 5 marks

Linked List

What is the issue?
temp = head
while temp.next != NULL:
print(temp.data)
temp = temp.next

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 temp = head
2 while temp != None:
3     print(temp.data)
4     temp = temp.next

```

Test Input

stdin input (optional)

Output

DEBUGGING & OPTIMIZATION TASK (Set 2)



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Q1 Binary Search 5 marks

Q2 Stack 5 marks

Q3 Stack 5 marks

Q4 Stack 5 marks

Q5 Stack 5 marks

Q6 Linked List 5 marks

Q7 Stack 5 marks

Q8 Linked List Stack 5 marks

Stack

Identify the errors in the following snippet.

```

if(ch == ')') {
    while(stack[top] != ')') {
        postfix[j++] = pop();
    }
    pop();
}

```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```

1 if (ch == ')') {
2     while (top != -1 && stack[top] != '(') {
3         postfix[j++] = pop();
4     }
5     if (top != -1 && stack[top] == '(') {
6         pop();
7     }
8 }

```

Test Input

stdin input (optional)

Output

Q3 Stack

5 marks

- Q1 Binary Search 5 marks ✓
- Q2 Stack 5 marks ✓
- Q3 Stack 5 marks ✓
- Q4 Stack 5 marks ✓
- Q5 Stack 5 marks ✓
- Q6 Linked List 5 marks ✓
- Q7 Stack 5 marks ✓
- Q8 ✓

Check the snippet and identify the logical error.

```
int stack[5];
int top;
void init() {
    top = 0;
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int stack[5];
2 int top;
3
4 void init() {
5     top = -1;
6 }
```

Test Input

stdin input (optional)

Output

Q4 Stack

5 marks

- Q1 Binary Search 5 marks ✓
- Q2 Stack 5 marks ✓
- Q3 Stack 5 marks ✓
- Q4 Stack 5 marks ✓
- Q5 Stack 5 marks ✓
- Q6 Linked List 5 marks ✓
- Q7 Stack 5 marks ✓
- Q8 ✓

Find the error in the stack deletion operation. Justify it.

```
int pop() {
    if(top == 0) {
        printf("Underflow");
        return -1;
    }
    return stack[top--];
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 int pop() {
2     if(top == -1) {
3         printf("Underflow");
4         return -1;
5     }
6     return stack[top--];
7 }
```

Test Input

stdin input (optional)

Output

Q3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marksQ8
Linked List Data Structures
5 marksQ9
Linked List
5 marks

Q10

return Q[front++]

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 DEQUEUE(Q):  
2 if front == -1:  
3 print("Underflow")  
4 else:  
5 return Q[front++]
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

QUESTIONS

Q1
Stack Notations - Postfix
Expression
5 marksQ2
Circular Queue
5 marksQ3
Binary Search
5 marksQ4
Queue
5 marksQ5
Queue
5 marksQ6
Doubly Linked List
5 marksQ7
Stack
5 marks

Q5 Queue

5 marks

Identify the error in the Queue enqueue logic.

```
void enqueue(int data) {  
    if (rear == MAX) return;  
    queue[rear++] = data;  
}
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 #define MAX 100  
2  
3 int queue[MAX];  
4 int front = -1, rear = -1;  
5  
6 void enqueue(int data) {  
7     if (rear >= MAX - 1) {  
8         printf("Queue is full\n");  
9         return;  
10    }  
11    if (front == -1) {  
12        front = 0;  
13    }
```

Test Input

stdin input (optional)

Output

Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 new.next = head
2 head = new
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

Stack
5 marks

Q3
Stack
5 marks

Q4
Stack
5 marks

Q5
Stack
5 marks

Q6
Linked List
5 marks

Q7
Stack
5 marks

Q8
Linked List Data Structures
5 marks

Q9
Linked List
5 marks

```
prev = current
current = current.next
```

Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 while current != NULL:
2     next_node = current.next
3     current.next = prev
4     prev = current
5     current = next_node
```

Test Input

stdin input (optional)

Output

Run

Save

Submit Fix

Q7 Stack

5 marks

- Q1 Binary Search
5 marks ✓
- Q2 Stack
5 marks ✓
- Q3 Stack
5 marks ✓
- Q4 Stack
5 marks ✓
- Q5 Stack
5 marks ✓
- Q6 Linked List
5 marks ✓
- Q7 Stack
5 marks ✓
- Q8 ✓

Point out the error.

```
POP(stack):  
if top == 0:  
    print("Underflow")  
else:  
    return stack[top--]
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 def POP(stack, top):  
2     if top == -1:  
3         print("Underflow")  
4         return None  
5     else:  
6         element = stack[top]  
7         top -= 1  
8         return element
```

Test Input

stdin input (optional)

Output

Q8 Linked List

5 marks

- Q1 Binary Search
5 marks ✓
- Q2 Stack
5 marks ✓
- Q3 Stack
5 marks ✓
- Q4 Stack
5 marks ✓
- Q5 Stack
5 marks ✓
- Q6 Linked List
5 marks ✓
- Q7 Stack
5 marks ✓
- Q8 ✓

What is the issue?

```
temp = head  
while temp.next != NULL:  
    print(temp.data)  
temp = temp.next
```

✳ Debugging Task: Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 temp = head  
2 while temp != None:  
3     print(temp.data)  
4     temp = temp.next
```

Test Input

stdin input (optional)

Output

Stack

5 marks

Q1
Binary Search
5 marksQ2
Stack
5 marksQ3
Stack
5 marksQ4
Stack
5 marksQ5
Stack
5 marksQ6
Linked List
5 marksQ7
Stack
5 marks

Q8

In balancing Bracket, you identify the error.

```
if((ch == '[' && stack[top] == '[') ||  
(ch == ']' && stack[top] == '[')) {  
    pop();  
} else {  
    return 0;  
}
```

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 if ((ch == '[' && stack[top] == '[') ||  
2     (ch == ']' && stack[top] == '[') ||  
3     (ch == '[' && stack[top] == '[')) {  
4     pop();  
5 } else {  
6     return 0;  
7 }
```

Test Input

stdin input (optional)

Output